

		<table><tr><td>D</td><td>650</td><td>500</td><td>3.1</td></tr></table>	D	650	500	3.1
D	650	500	3.1			
20	C	gain in kinetic energy = loss in electric potential energy $= qV_{\text{initial}} - qV_{\text{final}}$ $= qE\left(d - \frac{d}{4}\right) - qE\left(d - \frac{3d}{4}\right)$ $= \frac{1}{2}qEd$ $= \frac{1}{2}q\left(\frac{V}{d}\right)d$ $= \frac{1}{2}qV$ The gain in kinetic energy of the positive ion is independent of d.				
21	C					